

Allium Leaf Miner on Track to Emerge in Early April

We are approaching the time when allium leafminer will become active in PA and growers need to start now to prepare for this pest.

Updated: March 28, 2025



Allium leafminer on leek leaf. Photo: T. Elkner, Penn State Extension.

By now, most allium growers are familiar with Allium leaf miner (ALM), but I do still receive calls where this insect is being seen for the first time in a crop. ALM is an invasive fly from Poland that attacks plants in the Allium genus, including onion, garlic, leek, scallions, shallots, and chives. It overwinters as a pupa in leaf tissue or nearby soil, emerges in the spring, and adult flight occurs over a 5-7-week period. Females puncture leaves with their ovipositor, and both males and females feed on leaf sap. Oviposition results in a characteristic linear series of round wounds (Image 1).



(https://extension.psu.edu/media/wysiwyg/extensions/catalog_product/739e66be581c44d88f38ca2d6dd6a307/a//allium-leaf-miner.jpg)

Image 1. Leaf miner fly with oviposition and feeding scars on scallion. Photo: T. Elkner.

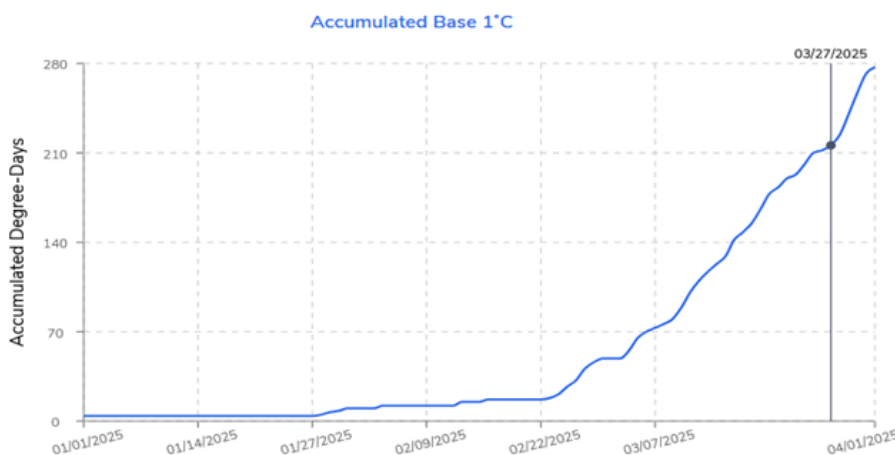
Larval development progresses to the pupal stage but is then delayed as the pupa undergoes summer aestivation or rest, and adults do not emerge again until late September for another 5-7-week flight.

The most important step in managing this insect is knowing when adults start flying. You can protect crops by applying netting or row covers prior to flight and removing them after the flight-period ends. Another option is to apply insecticides during the flight period, targeting adults and developing larvae. Systemic insecticides – with an added surfactant - tend to work best because the larvae are mining inside the leaf tissue. Since allium leaves are very waxy, a surfactant is recommended when insecticides are applied to allium crops. Results from our research trials found that the highest and most consistent control of ALM occurred using foliar applications of dinotefuran (Scorpion), cyantraniliprole (Exirel) and spinetoram (Radiant), and spinosad (Entrust) among OMRI-labeled options. Those trials used weekly applications starting as soon as oviposition/feeding marks were detected. More recent trials suggest that very good control can be achieved by waiting about 2 weeks after the first detection.

But when to start? With the support of PVGA and USDA, Dr. Shelby Fleischer from Penn State developed a degree day model. The model estimates that spring ALM emergence occurs after 350 degrees Celsius days above a threshold of 1°C, starting on January 1. It's a statistical estimate from field data and I have seen some flies on leaves before the model predicted emergence. However, my experience with the model has found that it is rather accurate for predicting the first heavy emergence of flies. If you are looking for 'indicator plants' rather than a model calculation, use the timing of when daffodils and forsythia bloom for a week. Another option is when ornamental pears are in bloom in urban areas. A third option is to use the 1-800-PENN-IPM (736-6476) phone hotline to listen to the messages I record. You need a touchtone phone to select and listen to these prerecorded messages. After dialing in, you select "1" for vegetables and then "2" for onions and alliums. I update the recording weekly during spring and fall when ALM are active based on scouting a sentinel plot at the Penn State research farm in Manheim as well as scouting in some commercial grower fields.

You can use the Northeast Weather Associate to estimate when emergence occurs by selecting a nearby weather station. Go to NEWA, click 'weather tools,' click 'Degree Day calculator,' pick a site near your farm, insert a start date of January 1 and end date of today, and insert degree day type as 1°C. The degree day selection is very important because you need to accumulate in C instead of F for the 350 threshold. If you use F, the conversion will mean we need ~650 degree days (dd). You get a chart and below that a cumulative graph. Figure 1 is the graph for Manheim on March 27, 2025. We are predicted to reach 277 degree days on April 1. Compared to last year's data from Bainbridge (the Manheim station was not active), we are behind 46 dd - last year, we had 323 degree days on April 1, with this year's predicted value at 287. These numbers lead me to predict that allium leaf miners will first appear in Lancaster County in the second week of April (barring any drastic changes in the weather – warmer or cooler than normal).

ALM has been observed in southern New Jersey, and it will soon (in southeast PA and within a week in cooler areas) be the time to start scouting your Allium crops and prepare for management. Growers planning to use row covers to exclude the adult flies should get them out of storage and have them put in place by the end of the month or by the middle of next week at the latest. When scouting, finding adults is easiest in the cool temperatures of early morning and looking at the tops of the leaves. I find that seeing the feeding scars on leaves is often easier than finding adults, especially on onions, and when scouting during windy conditions. Among Allium species, I tend to find more ALM on early spring scallions and green onions, so that is also a great place to search. Garlic, with its curling leaves, is more challenging to thoroughly scout, especially when trying to detect the earliest flies. You can also scout wild garlic in your fields and fencerows for leaf damage – I find it much easier to see leaf damage on wild rather than conventional garlic, but it always seems to be a secondary choice for flies to feed on.



(https://extension.psu.edu/media/wysiwyg/extensions/catalog_product/628bbfe63e9f44f3925ed720295ea7b5/p/i/picture1-png.png)

Figure 1: Accumulated Celsius degree days above a base threshold of 1°C, at Manheim, PA on March 27, 2025. ALM emergence is predicted to start at 350°C Degree days.

